

LA ROCHELLE ILE DE RE AP, France

WMO: 073160

Lat: 46.1781N

Lon: 1.1931W

Elev: 74

StdP: 14.66

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.8	29.6	20.4	15.4	28.7	23.8	18.0	31.8	30.6	49.2	27.4	49.8	9.0	40	0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	86.9	67.9	82.6	66.8	78.7	65.8	70.6	81.5	69.1	78.2	67.8	75.8	11.3	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.3	100.7	73.9	66.1	96.3	72.6	64.8	92.2	71.5	34.4	82.2	33.2	78.2	32.1	75.9	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.3	22.2	19.9	DB	23.3	94.5	4.0	4.5	20.5	97.8	18.1	100.4	15.9	102.9	13.0	106.2
			WB	22.0	73.4	4.1	1.8	19.1	74.7	16.7	75.7	14.4	76.7	11.4	78.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.7	44.0	44.3	48.9	53.4	58.7	64.9	68.6	67.3	64.0	58.6	50.3	45.1
	DBStd	10.48	6.49	6.57	5.36	4.86	4.92	5.81	5.05	4.72	5.34	5.70	6.14	6.75
	HDD50	730	200	170	83	18	2	0	0	0	0	7	71	179
	HDD65	3735	652	579	499	348	205	67	18	25	78	206	440	618
	CDD50	2825	12	11	49	121	270	449	576	536	421	274	81	25
	CDD65	357	0	0	0	1	8	66	129	96	49	8	0	0
	CDH74	2162	0	0	0	14	70	500	770	490	299	19	0	0
	CDH80	672	0	0	0	1	10	162	267	155	75	2	0	0
(14) Wind	WSAvg	10.4	10.4	11.1	11.2	10.3	10.6	10.3	10.5	10.2	9.7	9.6	10.7	10.8
(15) Precipitation	PrecAvg	28.9	2.9	2.2	2.0	2.4	2.0	1.4	1.7	1.9	2.2	3.3	3.8	3.1
	PrecMax	40.0	6.5	4.6	6.4	9.4	3.5	3.3	5.1	4.3	8.3	6.8	10.2	5.8
	PrecMin	20.6	0.4	0.0	0.2	0.6	0.6	0.2	0.2	0.2	0.0	0.4	0.7	0.6
	PrecStd	5.2	1.6	1.3	1.4	1.8	0.8	0.7	0.9	1.3	1.9	1.7	2.0	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.1	61.1	70.3	76.3	80.9	91.9	93.1	90.7	87.6	76.5	65.8	58.5
		MCWB	53.9	52.2	56.6	60.7	65.5	70.8	70.9	68.7	67.4	64.0	59.3	54.9
	2%	DB	55.4	56.4	64.3	70.1	75.0	84.5	86.8	84.3	81.3	72.0	62.8	56.7
		MCWB	53.4	50.8	54.5	58.2	62.8	67.2	68.3	67.4	65.4	63.2	58.4	53.6
	5%	DB	53.9	54.1	60.2	66.0	70.7	79.3	82.3	79.2	76.9	69.0	60.9	55.7
		MCWB	52.0	50.4	52.2	56.3	61.1	65.7	67.1	66.6	63.7	62.0	57.4	53.2
	10%	DB	52.6	52.6	56.7	62.4	67.4	75.4	77.5	75.1	73.3	66.5	59.3	54.5
		MCWB	50.7	49.7	51.2	55.0	59.6	64.4	66.5	65.8	63.8	61.2	56.1	52.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.2	54.3	58.0	62.8	68.1	72.5	72.5	71.8	70.8	66.3	61.0	56.3
		MCDB	55.8	58.1	67.4	72.5	78.6	86.5	85.4	82.9	80.1	72.2	63.4	57.3
	2%	WB	53.8	52.7	55.2	59.8	64.5	69.6	70.5	69.9	68.2	64.6	59.4	54.9
		MCDB	54.7	54.9	62.1	67.7	71.8	80.3	81.9	79.1	75.8	68.9	61.8	55.9
	5%	WB	52.5	51.4	53.7	57.7	62.3	67.6	69.0	68.3	66.5	63.4	58.1	53.7
		MCDB	53.6	53.4	57.8	64.5	68.5	76.3	78.6	75.5	72.8	67.4	60.5	55.2
	10%	WB	51.0	50.1	52.3	55.7	60.6	65.6	67.7	67.0	65.0	62.0	56.4	52.2
		MCDB	52.4	52.4	55.6	61.2	65.9	73.0	76.3	73.3	71.0	65.9	59.1	54.2
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	11.1	13.0	14.7	14.3	15.8	14.6	15.1	16.6	13.4	11.2	9.9
		MCDBR	8.4	13.3	19.5	21.9	20.9	23.9	24.0	23.1	22.7	16.0	11.7	9.2
	5% WB	MCWBR	7.3	9.6	11.7	12.7	11.4	10.7	10.2	10.1	10.5	9.1	8.1	7.5
		MCWBR	7.4	10.7	14.7	18.8	17.2	20.3	19.0	17.5	16.8	12.8	9.8	8.0
(40) Clear-Sky Solar Irradiance	taub		0.330	0.344	0.372	0.393	0.398	0.409	0.400	0.390	0.369	0.367	0.342	0.325
		taud	2.459	2.412	2.338	2.285	2.325	2.326	2.350	2.393	2.431	2.442	2.451	2.444
	Ebn at Noon		243	262	270	274	276	272	273	272	268	250	235	230
		Edn at Noon	22	28	34	39	39	39	38	35	31	27	22	20
(44) All-Sky Solar Radiation	RadAvg		383	687	1062	1507	1767	1979	1912	1693	1343	806	464	339
	RadStd		38	70	118	144	136	152	93	121	88	68	49	41

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page